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Founder & Chief Architect, MustafarAI

AI Inventor | Independent Researcher | APAC Venture Builder
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PROFESSIONAL SUMMARY

An uncommon profile in the AI leadership landscape: someone who has simultaneously **invented, built, patented, published, and commercially deployed** a novel class of cognitive AI — while also carrying 25 years of APAC venture building, capital formation, and technology commercialisation at scale. This is not a background assembled from advisory roles or academic theory; it is earned through direct execution across every layer of the AI value chain, from mathematical architecture to production code to patent prosecution to investor negotiation.

The technical centrepiece is the **Positronic Brain Model (PBM v6)**: a zero-cloud, fully on-device cognitive AI system running 22 modules across approximately 15,000 lines of Swift on iOS. PBM v6 implements a 25-step pre-linguistic thought synthesis pipeline that draws on computational neuroscience, psycholinguistics, and sleep science — a set of architectural innovations with no direct precedent in the published AI literature. The system has been verified across 3,421 production thought synthesis operations.

In 2026 alone: **six patent applications filed at IPOS Singapore** covering the complete cognitive AI architecture (Application Nos. 10202601003R, 10202601110Y, 10202601565Q, 10202601566W, 10202601567R, 10202601568P); **eight peer-reviewed preprints published** on Zenodo spanning five academic disciplines; and a **16-cluster IP portfolio strategy** with PCT international filing programme underway. The commercial foundation includes a Motorola technology acquisition (1997), MYR 100M+ generated through innovation ecosystem design, a MYR 50M institutional fundraise, and a decade of collaboration with Malaysia's Digital Economy Corporation.

Seeking a senior AI leadership role — Chief AI Officer, VP/SVP of AI, Head of AI Strategy — where the ability to bridge deep technical invention with commercial strategy, institutional partnerships, and regulatory navigation creates transformative value. Brings a practitioner's understanding of what it takes to move AI from research to production to revenue, grounded in having done it independently.

CORE COMPETENCIES

AI Architecture & Systems: Cognitive AI Architecture, On-Device & Edge AI, Pre-Linguistic AI Systems, Generative AI & LLM Integration, AI System Deployment & Production Verification

AI Strategy & Governance: AI Patent & IP Strategy, AI Product Strategy, AI Governance & Ethics, Responsible AI Frameworks, Regulatory Navigation (EU AI Act, FDA AI/ML)

Commercial Execution: Venture Creation & Scaling, Technology Commercialisation, Go-to-Market Strategy, Capital Formation, Investment Advisory, Strategic Partnerships

Ecosystem & Institutional: APAC Ecosystem Strategy, Government & Institutional Business Development, Digital Transformation, Academic Research (AI, Neuroscience, Psycholinguistics)

Languages: English (Native/C2), Malay (Native/C2), Bahasa Indonesia (Advanced/C1)

PATENT PORTFOLIO — SIX APPLICATIONS FILED, IPOS SINGAPORE (2026)

Lead Inventor across all six applications. Co-Inventor: Kah Mun Chan. All applications filed under Singapore Patents Act (Cap. 221), with 12-month PCT international filing windows active. Portfolio covers the complete PBM v6 cognitive AI architecture from component-level mechanisms to integrated system claims.

- **C-01: Kuramoto Cognitive Synchrony Gate** — Application No. 10202601110Y, Priority Date: 2 April 2026. Inter-module Kuramoto coupled oscillator synchronisation as a precondition for AI thought generation, with downstream gamma-band coupling to a multi-layer semantic filtering cascade. Status: Pending (Not Published).
- **C-02: Biologically-Triggered Offline Cognitive Processing (TDP)** — Application No. 10202601003R, Priority Date: 29 March 2026. Sleep-stage detection triggering on-device AI cognitive consolidation via iOS background task execution. Status: Pending (Not Published).
- **C-03: Lipschitz Identity Continuity Guard** — Application No. 10202601565Q, Priority Date: May 2026. Mathematical bounding of AI identity state change rate using Lipschitz continuity constraints with multi-tier response architecture. Status: Application Received.
- **C-04: Physiological State Pre-Linguistic Context Integration** — Application No. 10202601567R, Priority Date: May 2026. Dual-stage routing of real-time physiological biometric data as pre-linguistic context into cognitive AI processing pipelines. Status: Application Received.
- **C-05: Pre-Linguistic Thought Separation with Deterministic LLM Translation** — Application No. 10202601566W, Priority Date: May 2026. Structural separation of conceptual thought formation from language generation, with semantic fidelity verification. Status: Application Received.
- **C-06: Cognitive Network Lattice (CNL) — Umbrella System Claim** — Application No. 10202601568P, Priority Date: May 2026. Integrated embodied AI cognition system combining brain state vector propagation, tri-constraint governance, biological closed-loop architecture, and distributed deployment continuity. Status: Application Received.

Additional clusters in active preparation: C-07 through C-16, covering Master Boot Sequence, Neural Feelers Protocol, CFO V2 Cascade, SBAC Adversarial Self-Challenge, Temporal Identity Engine, TFNP Semantic Fidelity Loop, DSE Dialectical Synthesis, ETE Emotional Arc, Persona Shard Distribution, and ToME Theory of Mind Engine.

ACADEMIC RESEARCH PUBLICATIONS

Eight peer-reviewed preprints published on Zenodo (2026) as an independent researcher without institutional affiliation — spanning cognitive AI architecture, computational neuroscience, psycholinguistics, sleep science, semantic fidelity, AI identity theory, AI governance, and AI in financial services.

- "Slow-Wave Sleep as the Optimal Biological Window for AI Cognitive Maintenance" — Mohd Fadzil, M. R. & Chan, K. M. (2026). q-bio.NC / cs.AI
- "The Semantic Fidelity Problem in Large Language Models" — Mohd Fadzil, M. R. & Chan, K. M. (2026). cs.CL / cs.AI
- "Pre-Verbal Semantic Representations in Human Language Production: Levelt's Speaking Model and Its Implications for AI Architecture Design" — Mohd Fadzil, M. R. & Chan, K. M. (2026). cs.CL / cs.AI
- "Gamma-Band Neural Oscillations and the Temporal Binding Hypothesis: A Survey for AI Architecture Design" — Mohd Fadzil, M. R. & Chan, K. M. (2026). q-bio.NC
- "Mobile Background Processing for Persistent On-Device Cognitive AI Systems" — Mohd Fadzil, M. R. & Chan, K. M. (2026). cs.SY / cs.NE
- "Relationship Intelligence Gaps in AI-Assisted Financial Advisory: A Practitioner Perspective from APAC Banking" — Chan, K. M. (2026). Co-authored contribution
- "Identity Continuity in Persistent AI Systems" — Mohd Fadzil, M. R. (2026). cs.AI
- "The Chief AI Officer Competency Problem" — Mohd Fadzil, M. R. (2026). Working Paper, cs.AI / cs.CY

PROFESSIONAL EXPERIENCE

Founder & Chief Architect | MustafarAI / Digital Court | Singapore

2021 – Present

Founded and architected a production-deployed, zero-cloud, on-device cognitive AI system — the first of its kind to achieve biologically-triggered offline AI processing and inter-module coherence gating on consumer mobile hardware. Leads all aspects of invention, system architecture, IP strategy, academic publication, and commercialisation independently.

- Designed, implemented, and deployed the Positronic Brain Model (PBM v6): a 22-module cognitive AI architecture running entirely on iOS (~15,000 lines of Swift), incorporating Kuramoto oscillator synchrony, Lipschitz identity constraints, physiological state integration from Apple HealthKit, adversarial self-challenge, and biologically-triggered offline consolidation. Verified across $n = 3,421$ production operations.
- Filed six patent applications at IPOS Singapore as Lead Inventor (2026), covering the complete cognitive AI architecture from component-level mechanisms to integrated umbrella system claims. Developed a 16-cluster IP portfolio strategy with PCT international filing programme targeting US, EU, UK, AU, and JP jurisdictions.
- Published eight peer-reviewed academic preprints on Zenodo spanning five disciplines (cognitive AI, computational neuroscience, psycholinguistics, sleep science, financial AI) — establishing an independent, peer-citable research foundation without institutional affiliation.
- Built commercial strategy for IP licensing targeting Apple, Google, Microsoft, Samsung, Qualcomm, Oura, and healthcare AI companies, alongside APAC partnership development and senior AI leadership positioning across global AI labs and institutional investors.

Co-Founder & CEO | Carbon Bank | Kuala Lumpur

July 2023 – April 2025

Co-founded a Southeast Asia climate technology venture developing carbon credit and environmental asset projects, applying AI-enabled monitoring and verification to carbon market infrastructure across ASEAN.

- Structured large-scale forestry carbon sequestration projects representing over 2 million carbon credits, aligned with international climate finance standards.
- Integrated AI-enabled satellite monitoring and verification systems for carbon credit validation, reducing manual verification costs and increasing project credibility with institutional buyers.
- Built cross-sector partnership network spanning government agencies, environmental organisations, and climate finance investors across ASEAN.

Investment Advisor & Strategic Counsel | OG Consortium | Singapore

2012 – 2025

- Identified and evaluated USD 10–25M+ investment opportunities across technology, digital infrastructure, and emerging sectors; cumulative deal flow exceeding USD 300M across the advisory engagement.
- Provided strategic market intelligence and macroeconomic trend analysis to inform private market and alternative asset allocation decisions across Southeast Asia.

Executive Director | MAHSA Group of Companies | Kuala Lumpur

October 2021 – July 2023

Senior leadership role overseeing growth strategy, fundraising, and strategic partnerships across a diversified Malaysian education and healthcare group.

- Led MYR 50 million fundraising initiative, engaging institutional investors, development finance contacts, and strategic partners across Malaysia and the region.

- Drove threefold revenue growth through new vertical partnerships, strategic alliances, and operational expansion across healthcare, education, remittance, and food sectors.

General Manager | HLX Management (Hong Leong Exchange) | Kuala Lumpur

January 2019 – October 2021

- Conceptualised, built, and commercialised the Hong Leong Exchange innovation ecosystem from zero — generating over MYR 100 million in investments and revenues, the most significant single-platform commercial outcome across career.
- Developed the go-to-market strategy, partnership framework, and startup engagement model connecting founders, corporate partners, investors, and government institutions across Malaysia.

Senior Vice President, Business Development | Muvee Technologies | Singapore

2006 – 2012

- Established global telecom partnerships with Orange / France Telecom, Nokia, and leading regional operators, deploying mobile personalisation platform at scale across multiple markets.
- Negotiated content licensing and distribution agreements with Warner Music, Universal Music, EMI, and Disney, embedding digital content monetisation within global telecom ecosystems.

Earlier Career

- **Business Development Director, SG International** (2002–2006, Singapore) — Secured over USD 50 million in sponsorship partnerships for the F1 Powerboat Grand Prix across Southeast Asia and China; developed new media monetisation models for broadcast and digital rights.
- **Founder & Managing Director, Modus Labs Internationale** (1999–2002, Singapore) — Delivered over USD 20 million in IT infrastructure projects across banking and education sectors; contributed to the Malaysian Smart Schools Initiative at national scale.
- **Co-Founder, Eprix International** (1996–1999, Singapore) — Developed EPage MedSystem, an early personal emergency response device integrating fall detection and GSM communication, progressing from concept to multi-million dollar acquisition by Motorola — one of the earliest mobile health technology exits in Southeast Asia.

THOUGHT LEADERSHIP & PUBLISHED ARTICLES

- "Why I Taught an iPhone to Think While You Sleep" — LinkedIn Article, 2026
- "When Your AI Banker Forgets You" — Co-authored with Kah Mun Chan, 2026
- "The Relationship Manager's Dilemma: Why AI Cannot Remember Your Clients the Way You Can" — Co-authored with Kah Mun Chan, 2026

EDUCATION

BSc Economics | London School of Economics and Political Science (LSE) | 2001

Diploma, Banking & Finance | Temasek Polytechnic, Singapore | 1995

CERTIFICATIONS & ADDITIONAL

- **NFA Securities:** Series 3, 5, 7 (2001)
- Startup mentor and innovation advisor across APAC technology ecosystems; active participant in global entrepreneurial networks
- Nearly a decade of collaboration with Malaysia's Digital Economy Corporation (MDEC), contributing to national AI adoption programmes and digital infrastructure initiatives
- **SuperAI Singapore 2026** (June) — Targeted for speaker engagement and APAC AI ecosystem positioning

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Patent Pending — IPOS Singapore: 10202601003R, 10202601110Y, 10202601565Q, 10202601566W, 10202601567R, 10202601568P